

ML-based Chronic Kidney Disease and Diabetes Prediction With Feature Effect Analysis Using SHAP

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Abstract— In this paper, we introduce a machine learning (ML)-based approach for Chronic Kidney Disease (CKD) and diabetes prediction and perform feature effect analysis through SHAP (SHapley Additive exPlanations). We utilize two publicly available clinical datasets and build five ML classifier models for the analysis. Among all models, CatBoost provides the best performance for both CKD and diabetes prediction. Our CatBoost model has an accuracy of 0.95, mean 10-fold cross-validation of 0.96, ROC-AUC score of 0.99, precision of 0.96, recall of 0.96, and F1-score of 0.96 for CKD. The CatBoost model shows an accuracy of 0.99, mean 10-fold cross-validation of 0.97, ROC-AUC score of 1.0, precision of 1.0, recall of 0.98, and F1-score of 0.99 for diabetes. We perform comprehensive feature effect analysis by computing SHAP values. This gives insights into the contribution of every feature to the model's predictions. The achieved results highlight that CatBoost is a robust choice for accurate and reliable predictions of CKD and diabetes. The findings of this study contribute to the corresponding field of ML approaches for medical diagnosis and illustrate the significance of feature effect analysis in understanding model predictions. With excellent results, our research has the potential to enhance the clinical decision-making process and improve patient outcomes.

Keywords—Chronic Kidney Disease (CKD), Diabetes, Machine learning, SHAP, Feature Effect Analysis

I. INTRODUCTION

Chronic Kidney Disease (CKD) and Diabetes are affecting millions of people worldwide and have become global public health concerns. CKD shows rising incidence (over 800 million globally in 2023), and prevalence (13.4% globally). The associated mortality rate is also rising and caused 1.2 million deaths in 2023[1]. This has brought a significant load on healthcare systems, primarily in low-middle-income countries that have seen a high death rate due to the lack of proper renal replacement therapy [2]. Chronic kidney disease typically develops without early warning signs, and laboratory tests of the estimated glomerular filtration rate (eGFR) suggest irreversible kidney damage once 25% or more of it has been lost [3]. When it reaches the stage known as end-stage kidney disease, symptoms like leg swelling, extreme fatigue, weakness, shortness of breath, loss of appetite, or confusion arise [4]. If the underlying problem isn't controlled, the kidneys will continue toward irreversible deterioration. Underlying problems include high blood pressure, heart disease, obesity, and aging [5]. To prevent a rapid increase in death rates, hemodialysis or kidney transplant is critical to a medical condition for kidney failure. It is vital to discover kidney disease at its very first stage from

predictions, along with monitoring of CKD. To have preventive treatment on time and to better treat kidney disease. It can help the patient with a kidney disease popup. Preventions are in place to slow kidney damage and prolong life [6].

Patients with diabetes, characterized by abnormal blood glucose levels, face both immediate and long-term health hazards. If diabetes is not properly managed, it can promote cardiovascular disease, CKD, loss of vision, nerve damage, and death [7]. It is estimated that 10.5 percent of adults (537 million people) in the world between the ages of 20 and 80 have diabetes. Research has anticipated that this number will climb to 643 million by 2030 and 783 million by 2045 [8]. This diabetes will also cause nearly 1.5 million deaths per year. Effective diagnosis of diabetes ensures proper treatment and management. This can further help minimize the diabetes-associated risks and offer a better life quality to patients.

Machine Learning (ML) is being seen as a promising tool to develop effective diagnostics systems in the field of medicine. These ML systems depend upon algorithms that learn to classify people based on symptoms, distinguishing between folks who are unwell and healthy people. The application of ML is vital in uncovering hidden patterns and features for CKD and diabetes diagnosis [9]-[12]. This helps analyze the disease-related risk factors and promotes efficient early detection of diseases. Predictive models that are built with ML and DL algorithms, provide a possibility to triumph over the constraints of conventional disease-diagnosis methods. ML-based disease prediction can also improve the efficiency and sustainability of present healthcare practices. In this paper, we have analyzed two public datasets “Risk Factor prediction of Chronic Kidney Disease” [13] and “Early stage diabetes risk prediction” [14] from the UCI Machine Learning Repository [15]. Both of these datasets were donated within the past three years. We have developed multiple ML models to predict CKD and Diabetes using these clinical datasets. The main contribution of our paper is given below:

- Analyze the two public clinical datasets and build multiple ML models for CKD, and diabetes prediction.
- Validate and evaluate the models' performances by computing multiple performance metrics.
- Select the best-performing ML model for CKD and diabetes prediction by analyzing the models' performance metrics.

- Analyze the effect of features on the prediction model's decision-making process using SHAP (Shapley Additive exPlanations).

The remaining portions of the paper are arranged as follows. Section II highlights important research. Section III describes our methodology. Section IV shows our research results and discusses important findings. The paper is concluded in Section V.

II. RELATED WORKS

Many studies have been conducted to predict and diagnose CKD and diabetes using ML models, widely recognized for their high performance in the biomedical field.

A. CKD-based Existing Research

Machine learning techniques have proven valuable in monitoring and diagnosing chronic diseases, with a particular focus on CKD. The findings of these studies showed different uses of machine learning for the diagnosis and prediction of CKD in accurate and efficient ways, from traditional algorithms to deep learning models. In one study, seven machine learning algorithms, including ANN and linear SVM, were implemented to predict CKD using a dataset from the "UCI machine learning repository" [16]. Feature selection methods were applied, and the highest accuracy of 98.46% was achieved using LSVM. Chen et al. developed the KNN, SVM, and SIMCA models to predict patients' CKD risk using the UCI dataset [17]. They achieved the highest accuracy of 99.7% with SVM and KNN models. Six classification algorithms, including Naive Bayes, MLP, SVM, J48, and Decision Tree, were used to assess the accuracy and effectiveness of CKD classification. MLP provided the highest accuracy of 99.75% [18]. The DL algorithms, such as CNN, were applied to sensor datasets to detect CKD, achieving a prediction accuracy of 96.59%. A deep CNN was utilized to distinguish serum potassium levels in CKD patients, achieving high sensitivity (90%) with an AUC between 0.853 and 0.90. One DNN-based classifier attained an accuracy level of 98.25% in predicting CKD and its severity degree using the Particle Swarm Optimization (PSO) feature selection method [5], [19]. For CKD prediction, Boukenze et al. built various learning algorithms to enhance the accuracy of prediction. It included SVM, MLP, Decision Trees (C4.5), Bayesian Networks, and K-Nearest Neighbors. The authors compared models' efficiencies based on multiple criteria. They have found that the MLP and C4.5 Decision Trees have the highest accuracy rates [20].

B. Diabetes-based Existing Research

A technique for ensemble learning for early-stage diabetes prediction was presented in [21]. Using 17 attributes per record, the authors analyzed the UCI diabetes dataset. To choose features, Chi-2 was used. They used AdaBoost, bagging, and Random Forest (RF) as our three predictive models. The RF model outperformed other models significantly, as evidenced by the experimental findings, which demonstrated an excellent accuracy of 97%. Another research aimed at identifying type 2 diabetes and developed an ensemble method based on deep learning [22]. They had done evaluations using independent and ensemble deep learning models. The results by the authors

recommend the ensemble model CNN-BiLSTM, which has gained an accuracy of up to 88.33%. The investigation further contributes to the current literature demonstrating the merit of ensemble methods and deep learning techniques in enhancing the accuracy of early diabetes prediction models. A specialized deep neural network (DNN) model was designed to be used in classifying type-2 diabetes [23]. The model's design consists of performing stacked encoders for feature learning, applying backpropagation for network fine-tuning, and using the SoftMax function for classification. The model was then trained using 786 patient records in the Pima Indians Diabetes (PID) dataset. The results showed a classification accuracy of 86.26%, which justified the fact that the model effectively distinguishes diabetes types.

C. Comparison with Our Work

The previous works in predicting CKD and diabetes are commendable, but much is yet to be done to increase the performance of the prediction systems. This paper aims to present effective models that can perform prediction with high and reliable accuracy. We validate each model's efficiency using multiple performance metrics. We have also performed SHAP [24] analysis to understand how the features are influencing the model's prediction. Our analysis of these recent datasets will also help us understand the current scenario regarding the risk factors and symptoms of CKD and diabetes,

III. METHODOLOGY

In this paper, we study two public datasets and build ML models to predict kidney disease and diabetes. Our work is summarized in the flowchart given in Figure 1.

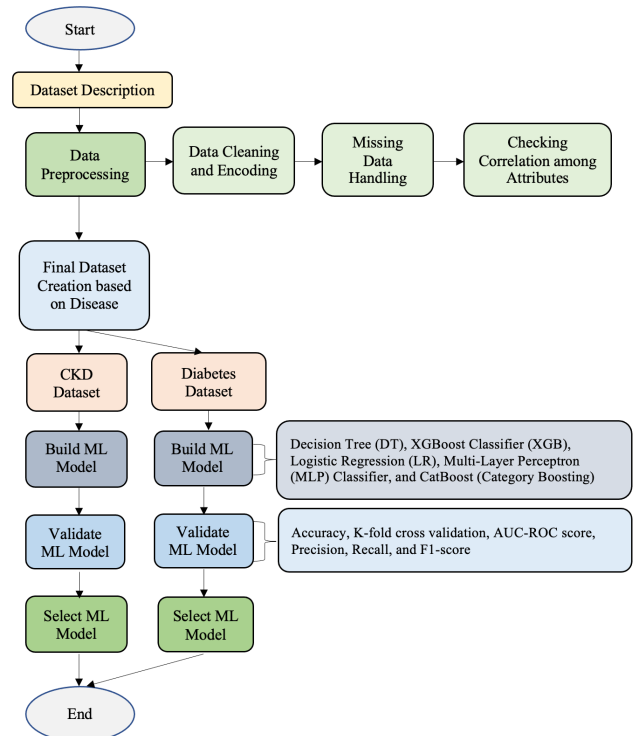


Fig. 1. Methodology Flowchart

A. Dataset Description

We acquire two public datasets from the UCI Machine Learning Repository. We use datasets titled “Risk Factor prediction of Chronic Kidney Disease” [13] and “Early stage diabetes risk prediction” [14] for CKD and diabetes prediction, respectively. We refer to “Risk Factor prediction of Chronic Kidney Disease” as the CKD dataset and “Early stage diabetes risk prediction” as the diabetes dataset in this paper. The CKD dataset has 202 instances and 29 features. All the feature types are real. The CKD dataset contains real patient data from Bangladesh. The data was collected in Enam Medical College, Savar, Dhaka, Bangladesh. The dataset contains Gold standard data of patients such as blood pressure, sugar level, albumin level, age, etc. The dataset was donated in August 2023. The diabetes dataset has 520 instances and 17 features. This was acquired through direct questionnaires. The subjects were the patients of Sylhet Diabetes Hospital in Sylhet, Bangladesh. The data collection was monitored and approved by a doctor. The dataset contains information about age, sex, polyuria, polydipsia, etc. The dataset was donated in July 2020.

B. Data Preprocessing

1) Data Cleaning and Encoding:

We convert the CSV files in “Pandas Dataframe” [25]. In the CKD dataset, there were two redundant rows. We deleted them. The attribute list of each dataset is shown in Table I.

TABLE I. DESCRIPTIONS OF DATASETS

Dataset	Attribute List	Type
CKD Dataset	'sg', 'al', 'su', 'bgr', 'bu', 'sod', 'sc', 'pot', 'hemo', 'pcv', 'rbcc', 'wbcc', 'grf', 'stage', 'age'	Categorical
	'bp (Diastolic)', 'bp limit', 'rbc', 'pc', 'pcc', 'ba', 'htn', 'dm', 'cad', 'appet', 'pe', 'ane', 'affected', 'class'	Integer
Diabetes Dataset	'Gender', 'Polyuria', 'Polydipsia', 'sudden weight loss', 'weakness', 'Polyphagia', 'Genital thrush', 'visual blurring', 'Itching', 'Irritability', 'delayed healing', 'partial paresis', 'muscle stiffness', 'Alopecia', 'Obesity', 'class'	Categorical
	'Age'	Integer

For categorical attributes, we check how many categories are in each attribute using “pandas.DataFrame.value_counts” [26]. Based on the total number of categories, we encode each categorical attribute to multiple classes such as 0, 1, 2, 3, etc. For the ‘class’ attribute in the CKD dataset, we encode ‘ckd’ and ‘notckd’ to 1 and 0, respectively. For the ‘class’ attribute in the diabetes dataset, we encode ‘Positive’ and ‘Negative’ to 1 and 0 respectively. For this data encoding, we have used “pandas.DataFrame.replace” [27]. For both datasets, ‘class’ represents the output column.

2) Missing Data Handling:

For both datasets, there was no missing data. For checking missing data, we have used “pandas.isnull” [28].

3) Checking Correlation among Attributes:

We have calculated the correlation among the attributes for both datasets using a Correlation matrix with Heatmap [29]. The Heatmap for CKD and Diabetes Dataset are shown in Figure 2 and Figure 3, respectively. From Figure 2, we see that the ‘affected’ and ‘class’ are totally correlated. That is why, we delete the ‘affected’ attribute from input.

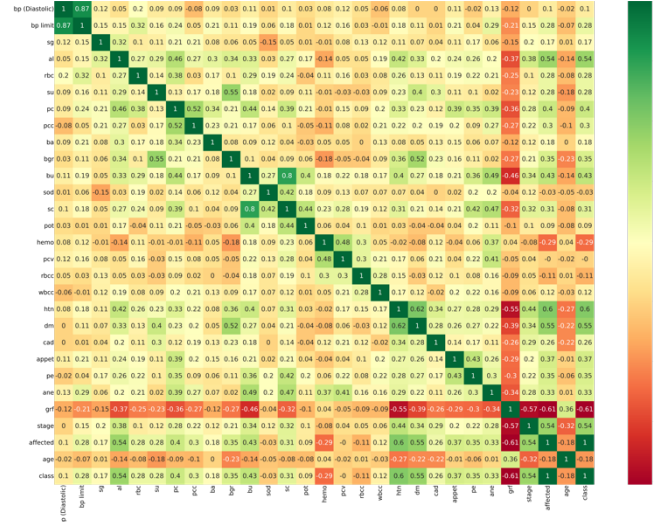


Fig. 2. Heatmap for CKD Dataset

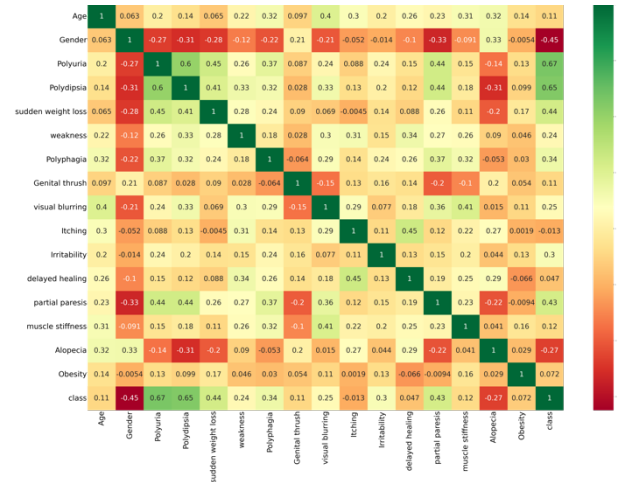


Fig. 3. Heatmap for Diabetes Dataset

C. Final Dataset Creation based on Disease:

Finally, we have a clean CKD dataset with 200 rows and 28 columns and a Diabetes dataset with 520 rows and 17 columns. The input and output for the CKD and Diabetes dataset are shown in Table II.

TABLE II. INPUT AND OUTPUT FOR CKD AND DIABETES DATASET

Dataset	Input	Output
CKD Dataset	'bp (Diastolic)', 'bp limit', 'sg', 'al', 'rbc', 'su', 'pc', 'pcc', 'ba', 'bgr', 'bu', 'sod', 'sc', 'pot', 'hemo', 'pcv', 'rbcc', 'wbcc', 'htn', 'dm', 'cad', 'appet', 'pe', 'ane', 'grf', 'stage', 'age'	'class'
Diabetes Dataset	'Gender', 'Polyuria', 'Polydipsia', 'sudden weight loss', 'weakness', 'Polyphagia', 'Genital thrush', 'visual blurring', 'Itching', 'Irritability', 'delayed healing', 'partial paresis', 'muscle stiffness', 'Alopecia', 'Obesity', 'Age'	'class'

After analyzing the output ('class') column of both datasets, we see that there are 128 subjects with CKD and 72 subjects without CKD in the CKD dataset. There are 320 subjects with diabetes and 200 subjects without diabetes in the Diabetes dataset. The class distribution for the CKD dataset and Diabetes dataset is shown in Figure 4 and Figure 5, respectively.

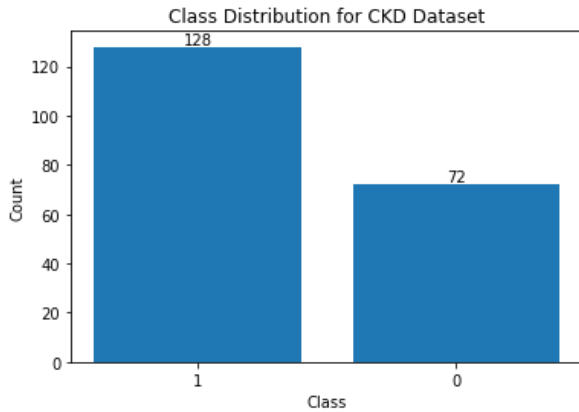


Fig. 4. CKD Dataset Class Distribution (1 = with CKD, 0 = no CKD)

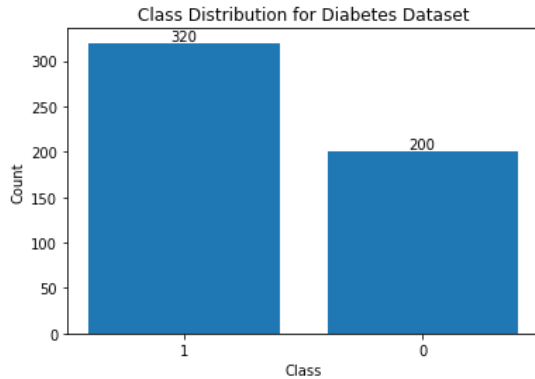


Fig. 5. Diabetes Dataset Class Distribution (1 = with Diabetes, 0 = no Diabetes)

D. Build ML Model:

Using "sklearn.model_selection.train_test_split" [30], we divide each dataset and use 80% data for training and 20% data for testing the model. For each dataset, we build five supervised ML models with Decision Tree (DT) [31], XGBoost Classifier (XGB) [32], Logistic Regression (LR) [33], Multi-Layer Perceptron (MLP) Neural Network [34], and CatBoost (Category Boosting) [35]. For developing models, we have used "sklearn.tree.DecisionTreeClassifier" [36], "xgboost" [37], "sklearn.linear_model.LogisticRegression" [38], "sklearn.neural_network.MLPClassifier" [39] and "CatBoostClassifier" packages [40].

E. Validate ML Model:

For validating the ML model's performance, we calculate accuracy, 10-fold cross-validation, ROC-AUC score, precision, recall, and F1-score [41]. Accuracy determines how accurate the model is performing. The 10-fold cross-validation helps minimize overfitting, shows trustworthy performance assessments, and guarantees the generalizability of the model. ROC-AUC indicates how well the model can differentiate between the two classes. Calculating Recall and precision are vital in situations where misclassifications can have adverse outcomes such as disease prediction. Precision and Recall balance false positives and false negatives. By combining recall and precision, the F1-score acts as a single metric for assessing classifier models. Equations for accuracy, precision, recall, and F1-score are given in Equation 1, Equation 2, Equation 3, and Equation 4, respectively. Together, all six metrics ensure the development of robust and reliable ML models.

$$Accuracy = \frac{\text{Total number of correct predictions}}{\text{Total number of predictions}} \quad (1)$$

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

$$Recall = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

$$F1 - score = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

IV. RESULT AND DISCUSSION

A. Result:

In this paper, we aim to predict CKD and diabetes by building ML models, leveraging two public datasets. We have employed Decision Tree (DT), XGBoost (XGB), Logistic Regression (LR), MLP Neural Network (MLP), and CatBoost classification algorithms for this purpose. The comprehensive summary of the performances of the developed ML models is given in Table III.

After comparing all models' performance metrics, we see that CatBoost predicts both CKD and diabetes with the highest accuracy. The accuracy is 0.95 for the CKD dataset and 0.99 for the diabetes dataset.

TABLE III. THE COMPREHENSIVE SUMMARY OF CKD AND DIABETES PREDICTION WITH DEVELOPED ML MODELS

Model Name	CKD Dataset						Diabetes Dataset					
	Accuracy	10-fold cross-validation	ROC-AUC Score	Precision	Recall	F1-score	Accuracy	10-fold cross-validation	ROC-AUC Score	Precision	Recall	F1-score
DT	0.90	0.86	0.91	0.95	0.87	0.91	0.96	0.96	0.96	0.98	0.95	0.968
XGB	0.95	0.945	0.99	0.96	0.958	0.96	0.95	0.93	0.99	0.94	0.98	0.96
LR	0.85	0.93	0.97	0.83	0.90	0.87	0.89	0.93	0.95	0.92	0.90	0.91
MLP	0.93	0.94	0.97	0.96	0.92	0.94	0.97	0.97	0.995	0.98	0.97	0.98
CatBoost	0.95	0.96	0.99	0.96	0.96	0.96	0.99	0.97	1.0	1.0	0.98	0.99

B. Discussion

1) Performance comparison of all models

Among the five developed models, CatBoost, XGB, MLP, and DT show accuracies of over 90% for both datasets. This indicates that these models are performing very well. CatBoost beats other models with the highest accuracy of 0.95 and 0.99, for CKD and diabetes prediction respectively.

2) Validation of ML models' performance

We validated each model's performance with our test data. We have calculated the average 10-fold cross-validation score for each model. The 10-fold cross-validation technique plays a vital role in robust model evaluation. The CatBoost shows very impressive average 10-fold cross-validation scores of 0.96 for CKD and 0.97 for the diabetes dataset.

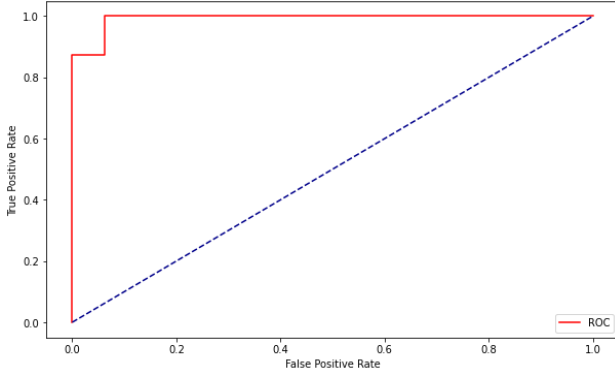


Fig. 6. ROC curve of CatBoost model for CKD Prediction

ROC-AUC determines a classifier model's ability to discriminate between classes. The score can range from 0-1. The better the score, the higher the discrimination capability the model has. This score also helps identify the model's performance in the case of an imbalanced dataset. Models with ROC-AUC score over 0.90 have outstanding discrimination competency [42]. All our models show a high score of over 0.90. The best model CatBoost has an ROC-AUC score of 0.99 for the CKD dataset and 1.0 for the diabetes dataset. This means

the model is accurately predicting CKD and diabetes. The ROC curve of CatBoost for CKD and Diabetes dataset is given in Figure 6 and Figure 7, respectively.

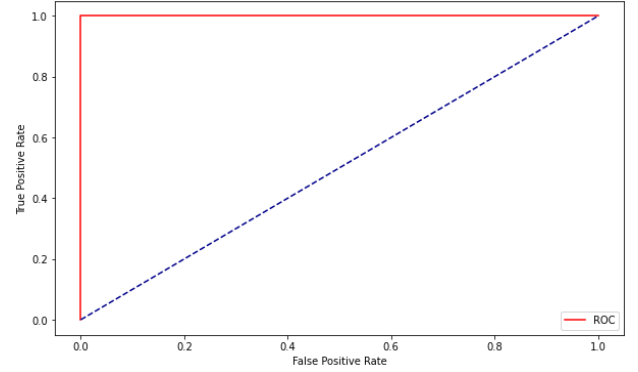


Fig. 7. ROC curve of CatBoost model for Diabetes Prediction

Precision, recall, and F1 score play a crucial role in predicting disease [43]. In disease prediction, false positives and false negatives lead to negative clinical implications. High precision indicates that the model is predicting the positive cases accurately. The model's high recall indicates that the model is effectively capturing high proportions of actual positive cases. F1-score balances between precision and recall and analyzes the model's performance comprehensively. Among all models, the Catboost shows the highest precision of 0.96, recall of 0.96, and F1-score of 0.96 for the CKD dataset. For the diabetes dataset, the Catboost has a precision of 1.0, a recall of 0.98, and an F1-score of 0.99. The high precision, recall, and an F1-score ensures that our CatBoost model performs accurate prediction. It also has the potential to facilitate early diagnosis. The confusion matrices [44] of the CatBoost model for the CKD and diabetes datasets are given in Figure 8 and Figure 9, respectively. This will provide more understanding of the model's performance with true positives, true negatives, false positives, and false negatives. Both of the confusion matrices show that the CatBoost is performing outstanding for both CKD and Diabetes prediction.

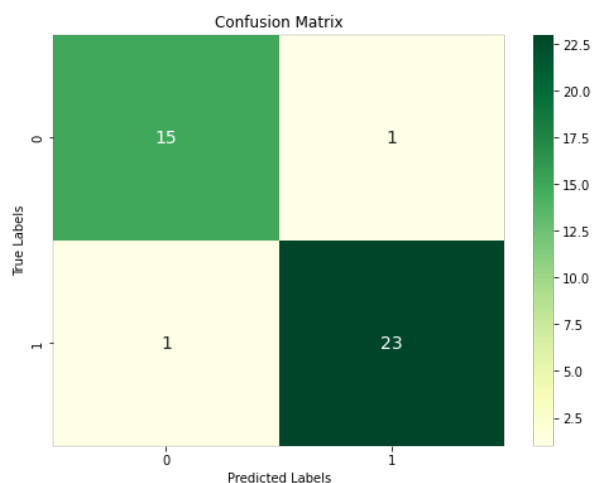


Fig. 8. Confusion matrix for CKD

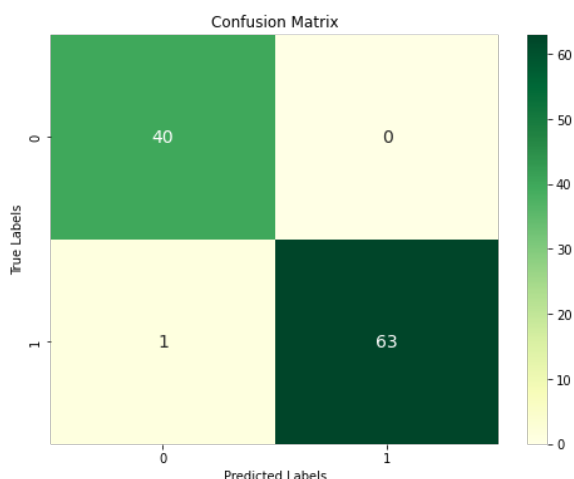


Fig. 9. Confusion matrix for Diabetes

3) Feature effect analysis on model

We have analyzed each feature's effect on the prediction of the ML model. We have calculated the SHAP (SHapley Additive exPlanations) [45] value for both datasets using the developed CatBoost model. SHAP values help us to mitigate the black-box nature of ML prediction models. It helps understand the contribution and impact of each feature in the model's prediction and decision-making process [46]. The SHAP summary plots with the CatBoost model for the CKD and diabetes dataset are given in Figure 10 and Figure 11.

In Figure 10 and Figure 11, the red dot to the right indicates the high value of that feature positively impacts prediction. The red dot to the left shows high feature value has a negative contribution to the prediction. The blue dot on the right side says that the low value of that feature enhances prediction. The blue dot on the left side says the low value of that feature negatively impacts prediction. In the SHAP plot (Figure 10 and Figure 11),

features at the top have higher absolute mean SHAP values. This indicates their higher importance on the model's prediction.

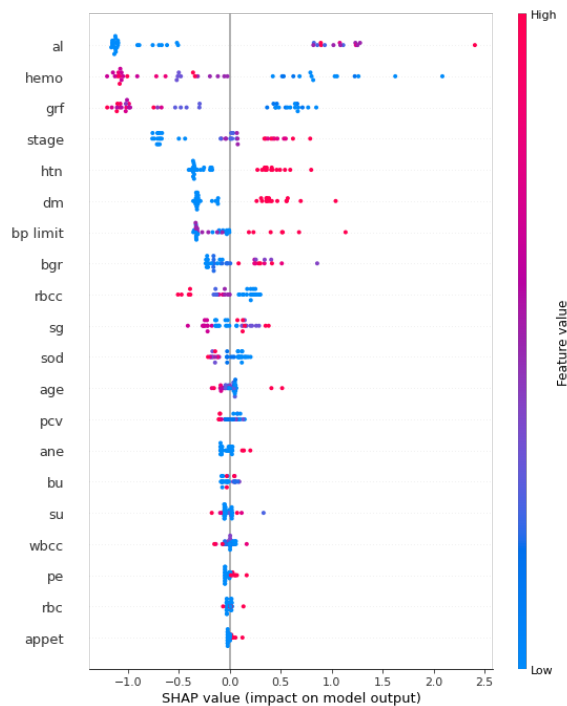


Fig. 10. SHAP summary plot for CKD prediction with CatBoost

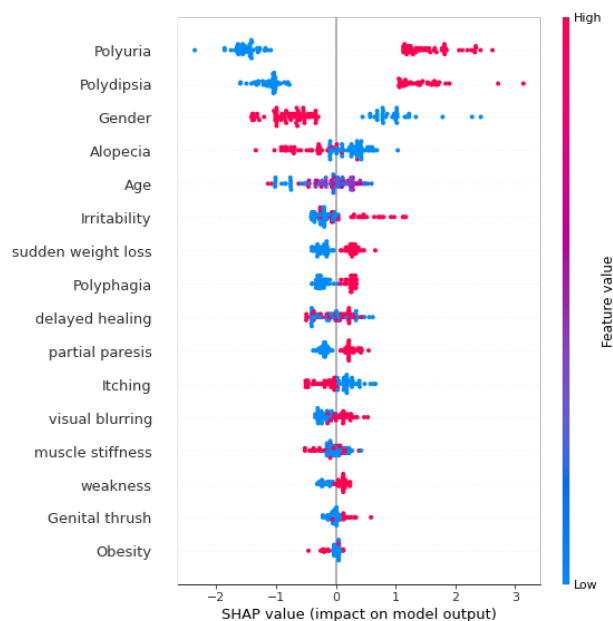


Fig. 11. SHAP summary plot for Diabetes prediction with CatBoost

V. CONCLUSION

In conclusion, we create five ML models for CKD and diabetes prediction, leveraging two public datasets. CatBoost outperforms all other models for both CKD and diabetes datasets and provides remarkable results in terms of accuracy, 10-fold cross-validation score, ROC-AUC score, precision, recall, and F1-score. We calculate SHAP values for feature

effect analysis. The exploration of SHAP values has highlighted how every feature contributes to the ML model's prediction process. This helps to understand the model's interpretability. Finally, the accurate and reliable models built in this research have the potential to perform faster, more accurate, and early diagnosis and create a notable impact in medical science.

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